WORKED SOLUTION

STEP III 2024, Question 9

Mechanics • Oblique Collisions

STEP III 2024, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Everything happens in the plane of the table, so look down from above: each sphere becomes a circle of radius r centred at its lowest point. For smooth spheres the impulse acts along the line of centres, so we split velocities into the component along that line (where momentum and restitution apply) and the component across it (which is unchanged). After that, B moves in a straight line, and whether it hits C is a question about the distance from a point to a line.

Solution

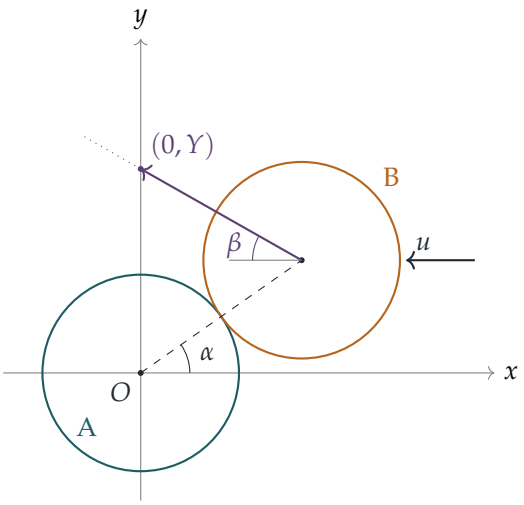
We model the spheres as smooth and equal in mass, on a smooth table, so there is no friction anywhere and each sphere moves in a straight line at constant speed except during the impact. We work with the lowest points, which move exactly as the centres do horizontally; two spheres of radius r touch when their lowest points are $2r$ apart.

Part (i)

B's lowest point moves along the line $y = 2r\sin\alpha$. It touches A when its distance from O is $2r$, that is at the point $(2r\cos\alpha, 2r\sin\alpha)$ (taking the positive x-coordinate, since B comes from large positive x). So at impact the line of centres, from A to B, has unit vector

$$\mathbf{n} = \begin{pmatrix}\cos\alpha\\ \sin\alpha\end{pmatrix}, \qquad \text{and} \qquad \mathbf{t} = \begin{pmatrix}-\sin\alpha\\ \cos\alpha\end{pmatrix}$$

is perpendicular to it.



Before impact B has velocity $\begin{pmatrix} -u \\ 0 \end{pmatrix}$, with components $-u\cos\alpha$ along $\mathbf{n}$ and $u\sin\alpha$ along $\mathbf{t}$. A is at rest.

The spheres are smooth, so the impulse between them is along $\mathbf{n}$. Hence B's $\mathbf{t}$-component stays $u\sin\alpha$ (and A gains no $\mathbf{t}$-component). Let the $\mathbf{n}$-components after impact be v_A and v_B. Conservation of momentum along $\mathbf{n}$ (equal masses) and Newton's law of restitution give

$$\begin{aligned} v_A + v_B &= -u\cos\alpha, \\ v_B - v_A &= \tfrac{1}{3}(0 - (-u\cos\alpha)) = \tfrac{1}{3}u\cos\alpha. \end{aligned}$$

(The approach speed along $\mathbf{n}$ is $u\cos\alpha$, and B, on the positive-$\mathbf{n}$ side, separates from A.) Adding, $v_B = -\frac{1}{3}u\cos\alpha$, and $v_A = -\frac{2}{3}u\cos\alpha$.

So B's velocity after impact is

$$\begin{aligned} -\tfrac{1}{3}u\cos\alpha \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} + u\sin\alpha \begin{pmatrix} -\sin\alpha \\ \cos\alpha \end{pmatrix} &= \begin{pmatrix} -\frac{1}{3}u(\cos^2\alpha + 3\sin^2\alpha) \\ \frac{2}{3}u\sin\alpha\cos\alpha \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{3}u(1 + 2\sin^2\alpha) \\ \frac{2}{3}u\sin\alpha\cos\alpha \end{pmatrix}, \end{aligned}$$

using $\cos^2\alpha = 1 - \sin^2\alpha$. This is the required result.

Part (ii)

Both components are nonzero, with the x-component negative and the y-component positive (as α is acute). So B now travels up and to the left, and the gradient of its path has size

$$\frac{\frac{2}{3}u\sin\alpha\cos\alpha}{\frac{1}{3}u(1 + 2\sin^2\alpha)} = \frac{2\sin\alpha\cos\alpha}{1 + 2\sin^2\alpha} = \tan\beta,$$

where β is the acute angle the path makes with the negative x-direction. B's lowest point starts at $(2r\cos\alpha, 2r\sin\alpha)$. To reach the y-axis it must decrease x by $2r\cos\alpha$, and meanwhile y increases by $2r\cos\alpha\tan\beta$. So it crosses the y-axis at $(0, Y)$ with

$$Y = 2r\sin\alpha + 2r\cos\alpha\tan\beta = 2r(\cos\alpha\tan\beta + \sin\alpha),$$

as required.

Part (iii)

After the impact B moves in a straight line at constant speed, along the line ℓ through $(0, Y)$ with unit direction $\mathbf{d} = (-\cos\beta, \sin\beta)$. C stays at rest at $(0, h)$ until something hits it. B strikes C exactly when the distance from $(0, h)$ to B's path drops below $2r$.

The perpendicular distance from $(0, h)$ to ℓ is the size of the component of $(0, h) - (0, Y) = (0, h - Y)$ perpendicular to $\mathbf{d}$. A unit normal to ℓ is $(\sin\beta, \cos\beta)$, so this distance is

$$|(0,\ h - Y)\cdot(\sin\beta, \cos\beta)| = |h - Y|\cos\beta,$$

since β is acute and so $\cos\beta > 0$. If $h > Y + 2r\sec\beta$, then $h - Y > 2r\sec\beta > 0$ and the distance is $(h - Y)\cos\beta > 2r$. Every point of B's path, not just the closest point of the whole line, is at least this far from C's lowest point. So B never comes within $2r$ of C and does not strike it.

(Nothing else gets in the way: A moves off with velocity $-\frac{2}{3}u\cos\alpha\,\mathbf{n}$, into $y < 0$, away from both B and C.)

> **Ask yourself** Part (iv) asks us to prove $Y < 2r\sec\beta$. Can we turn it into a single trigonometric fact?

Yes: multiply through by $\cos\beta > 0$ and the left-hand side becomes a compound-angle formula.

Part (iv)

Dividing by $2r$ and multiplying by $\cos\beta > 0$, the inequality $Y < 2r\sec\beta$ is equivalent to

$$\cos\alpha\sin\beta + \sin\alpha\cos\beta < 1, \qquad \text{that is,} \qquad \sin(\alpha + \beta) < 1.$$

Certainly $\sin(\alpha + \beta) \le 1$, with equality only if $\alpha + \beta = \frac{\pi}{2}$ (as $0 < \alpha + \beta < \pi$). That would mean $\tan\beta = \cot\alpha$, i.e.

$$\frac{2\sin\alpha\cos\alpha}{1 + 2\sin^2\alpha} = \frac{\cos\alpha}{\sin\alpha} \iff 2\sin^2\alpha = 1 + 2\sin^2\alpha$$

(dividing by $\cos\alpha \neq 0$), which is impossible. Hence $\sin(\alpha + \beta) < 1$, and so $Y < 2r\sec\beta$.

Hence. Now

$$Y + 2r\sec\beta < 4r\sec\beta,$$

so we need an upper bound on $\sec\beta$ over all acute α, i.e. an upper bound on $\tan\beta$. Dividing top and bottom by $\cos^2\alpha$ and writing $t = \tan\alpha > 0$,

$$\tan\beta = \frac{2\tan\alpha}{\sec^2\alpha + 2\tan^2\alpha} = \frac{2t}{1+3t^2} \le \frac{2t}{2\sqrt{3}\,t} = \frac{1}{\sqrt{3}},$$

by the AM–GM inequality $1 + 3t^2 \ge 2\sqrt{3}\,t$ (equality when $t = 1/\sqrt{3}$, i.e. $\alpha = \frac{\pi}{6}$). So $0 < \beta \le \frac{\pi}{6}$, which gives $\sec\beta \le \sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$. Therefore, for every acute α,

$$Y + 2r\sec\beta < 4r\sec\beta \le \frac{8r}{\sqrt{3}}.$$

If $h > \dfrac{8r}{\sqrt{3}}$ then $h > Y + 2r\sec\beta$, and by part (iii) B does not strike C, whatever the value of α.

Checks

- Momentum: A's velocity $-\frac{2}{3}u\cos\alpha\,(\cos\alpha, \sin\alpha)$ plus B's velocity from part (i) adds up to $(-u, 0)$, as it should for equal masses.
- When $\alpha = 0$ (a head-on impact) the formula gives B velocity $(-\frac{1}{3}u, 0)$ and A gets $(-\frac{2}{3}u, 0)$: the usual one-dimensional answer for equal masses with $e = \frac{1}{3}$.
- With $\alpha = \frac{\pi}{6}$ we get $\tan\beta = 1/\sqrt{3}$, so $\beta = \frac{\pi}{6}$, $\alpha + \beta = \frac{\pi}{3}$ and $Y = 2r(\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} + \frac{1}{2}) = 2r$, comfortably less than $2r\sec\beta = 4r/\sqrt{3} \approx 2.31r$. Then $Y + 2r\sec\beta \approx 4.31r$, below $8r/\sqrt{3} \approx 4.62r$.
- We confirmed by computer, for 20,000 random values of α, that stepping B's straight path gives the crossing point Y above, that B misses C just above $h = Y + 2r\sec\beta$ and hits it just below, and that the largest value of $\sec\beta$ is $1.1547\ldots = 2/\sqrt{3}$.

Common slips

- Applying restitution to the x-components instead of the components along the line of centres. The impulse between smooth spheres acts along the line of centres, which here is at angle α to the x-axis.
- Taking the line of centres at impact to be along the x-axis or the y-axis. The contact point is fixed by the spheres being $2r$ apart.
- In part (iii), testing whether C's lowest point lies within $2r$ of the single point $(0, Y)$ rather than of the whole path.

Beyond STEP

Splitting a velocity into components along and across a chosen unit vector is projection onto an orthonormal basis, an idea that runs through first-year linear algebra and

mechanics. Oblique collisions of smooth spheres are a first model of scattering, which physicists use to study gases and particle collisions. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions in two dimensions with this vector approach.

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